

Solution of Cauchy Problem for First Order PDEs

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Question

Solve $u(x+y)p + u(x-y)q = x^2 + y^2$, with Cauchy data $u = 0$ on $y = 2x$

The given equation is of the form
 $a(x,y,u)p + b(x,y,u)q = c(x,y,u)$, where

$$\begin{aligned} \rightarrow a(x, y, u) &:= u \cdot (x + y); \\ b(x, y, u) &:= u \cdot (x - y); \\ c(x, y, u) &:= x^2 + y^2; \end{aligned}$$

$$a(x, y, u) := u(x + y)$$

$$b(x, y, u) := u(x - y)$$

$$c(x, y, u) := x^2 + y^2$$

The characteristic equations are
 $dx/a(x,y,u) = dy/b(x,y,u) = du/c(x,y,u)$

1 Calculating ϕ

Consider the first two ratios

$$\rightarrow dx/a(x, y, u) = dy/b(x, y, u);$$

$$\frac{dx}{u(y+x)} = \frac{dy}{u(x-y)}$$

After simplification, we get
 $x dx - y dy - (y dx + x dy) = 0$
or $x dx - y dy - d(xy) = 0$
Integrating, we get

$$\rightarrow \text{integrate}(x, x) - \text{integrate}(y, y) - x \cdot y;$$

$$-\frac{y^2}{2} - xy + \frac{x^2}{2}$$

$$\rightarrow \phi : \% = c1; /* assigning the name \phi to the above expression = c1 */$$

$$-\frac{y^2}{2} - xy + \frac{x^2}{2} = c1$$

2 Calculating ψ

Now using the multipliers, $x, -y, -u$,
i.e. considering $(x dx - y dy - u du)/(x a - y b - u c)$,
the expression $(x a - y b - u c)$ becomes

$$\rightarrow \text{ratsimp}(x \cdot a(x, y, u) - y \cdot b(x, y, u) - u \cdot c(x, y, u));$$

Hence, $x dx - y dy - u du = 0$

Integrating, we get

--> $\text{integrate}(x, x) - \text{integrate}(y, y) - \text{integrate}(u, u);$

$$-\frac{y^2}{2} + \frac{x^2}{2} - \frac{u^2}{2}$$

--> $\psi : \% = c2;$

$$-\frac{y^2}{2} + \frac{x^2}{2} - \frac{u^2}{2} = c2$$

3 The General Solution $f(\phi, \psi) = 0$

The general solution is

--> $f(\text{lhs}(\phi), \text{lhs}(\psi)) = 0;$

$$f\left(-\frac{y^2}{2} - xy + \frac{x^2}{2}, -\frac{y^2}{2} + \frac{x^2}{2} - \frac{u^2}{2}\right) = 0$$

4 Using Cauchy Data

It is given that $u(x, 2x) = 0$

--> $\text{ev}(\phi, y = 2 \cdot x, u = 0);$
 $\text{ev}(\psi, y = 2 \cdot x, u = 0);$

$$-\frac{7x^2}{2} = c1$$

$$-\frac{3x^2}{2} = c2$$

We have

$$(2/7)c1 = (2/3)c2$$

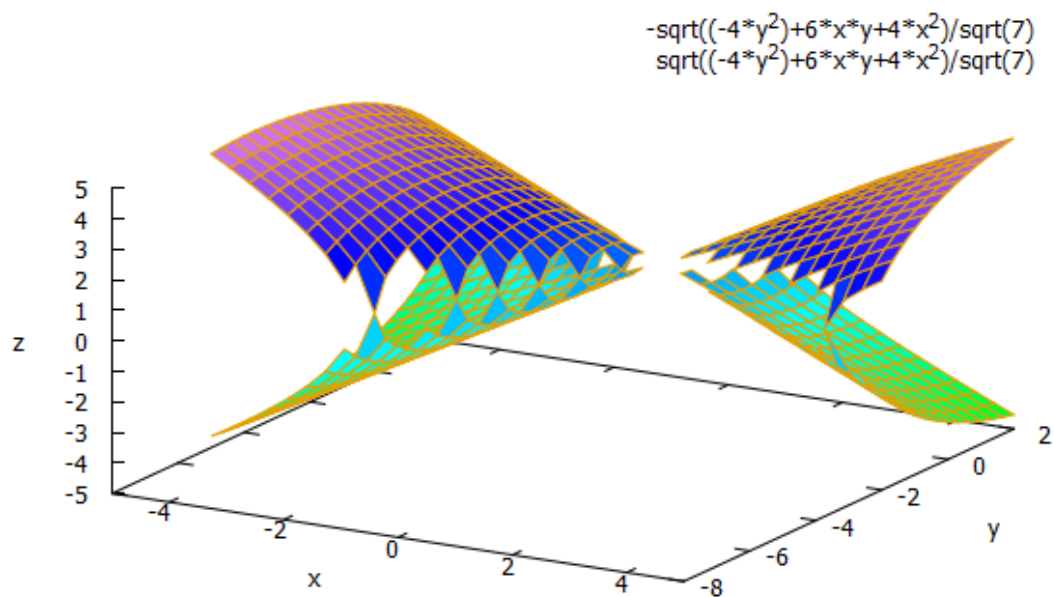
--> $\text{ratsimp}((2/7) \cdot \text{lhs}(\phi) - (2/3) \cdot \text{lhs}(\psi)) = 0;$

$$\frac{4y^2 - 6xy - 4x^2 + 7u^2}{21} = 0$$

5 Plotting the Solution

-- $\text{plot3d}([\text{sqrt}((4 \cdot x^2 - 4 \cdot y^2 + 6 \cdot x \cdot y)/7), -\text{sqrt}((4 \cdot x^2 - 4 \cdot y^2 + 6 \cdot x \cdot y)/7), [$
 > $x, -5, 5], [y, -5, 5]])$;

Figure 1:



```
--> kill ( all ) ;
```

done

Question

Solve $p - q = 1$, with Cauchy data $u(x,0) = x^2$

```
--> a ( x , y , u ) := 1 ;
      b ( x , y , u ) := - 1 ;
      c ( x , y , u ) := 1 ;
```

$$a(x, y, u) := 1$$

$$b(x, y, u) := -1$$

$$c(x, y, u) := 1$$

consider the first two ratios

Calculating ϕ

```
--> dx / a ( x , y , u ) = dy / b ( x , y , u ) ;
```

$$dx = -dy$$

after simplification we get $dx+dy=0$

integrating we get

```
--> integrate ( 1 , x ) + integrate ( 1 , y ) ;
```

$$y + x$$

```
-->  $\phi$  : % = c1 ;
```

$$y + x = c1$$

Calculating ψ

consider the last two ratios

$$\rightarrow dy/b(x, y, u) = du/c(x, y, u);$$

$$-dy = du$$

after simplification we get $du+dy=0$

$$\rightarrow \text{integrate}(1, u) + \text{integrate}(1, y);$$

$$y + u$$

$$\rightarrow \psi : \% = c2;$$

$$y + u = c2$$

The General Solution $f(\phi, \psi)=0$

the general solution is

$$\rightarrow f(\text{lhs}(\phi), \text{lhs}(\psi)) = 0;$$

$$f(y + x, y + u) = 0$$

Using Cauchy Data

it is given that $u(x,0)=x^2$

$$\rightarrow \begin{aligned} &\text{ev}(\phi, y=0, u=x^2); \\ &\text{ev}(\psi, y=0, u=x^2); \end{aligned}$$

$$x = c1$$

$$x^2 = c2$$

we have $c2=(c1)^2$ or $(c1)^2-c2=0$

$$\rightarrow \text{ratsimp}((\text{lhs}(\phi))^2 - \text{lhs}(\psi) = 0);$$

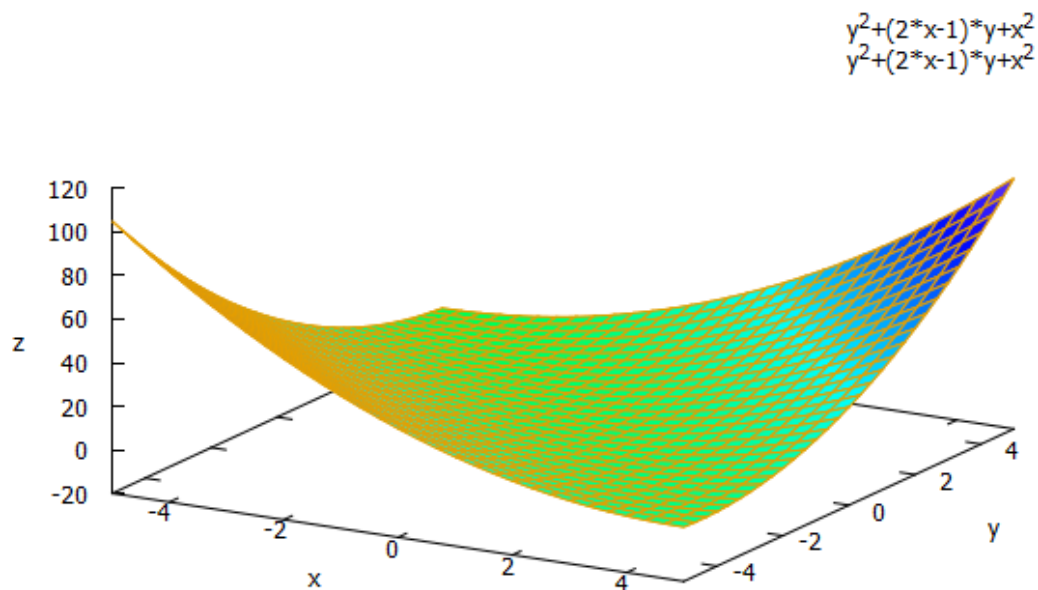
$$y^2 + (2x - 1)y + x^2 - u = 0$$

Plotting the Solution

$$\rightarrow \text{plot3d}([(y^2 + (2 \cdot x - 1) \cdot y + x^2), y^2 + (2 \cdot x - 1) \cdot y + x^2, [x, -5, 5], [y, -5, 5]]);$$

[\"C : /Users/Student/AppData/Local/Temp/maxout4936.gnuplot\"]

Figure 2:



Question

$(y-u)p + (u-x)q = x - y$, with Cauchy data $u = 0$ on $xy = 1, y = (x)^{-1}$

--> $a(x, y, u) := y - u$;
 $b(x, y, u) := u - x$;
 $c(x, y, u) := x - y$;

$$a(x, y, u) := y - u$$

$$b(x, y, u) := u - x$$

$$c(x, y, u) := x - y$$

consider the first two ratios

calculating Φ

using multipliers 1,1,1

we get $dx + dy + du = 0$

--> $\text{integrate}(1, x) + \text{integrate}(1, y) + \text{integrate}(1, u)$;

$$y + x + u$$

--> $\varphi : \% = c1$;

$$y + x + u = c1$$

Using multipliers x, y, u we get

$x dx + y dy + u du = 0$

--> $\text{integrate}(x, x) + \text{integrate}(y, y) + \text{integrate}(u, u)$;

$$\frac{y^2}{2} + \frac{x^2}{2} + \frac{u^2}{2}$$

--> $\psi : \% = c2 ;$

$$\frac{y^2}{2} + \frac{x^2}{2} + \frac{u^2}{2} = c2$$

now the general solution is

--> $f(\text{lhs}(\varphi), \text{lhs}(\psi)) = 0 ;$

$$f\left(y + x + u, \frac{y^2}{2} + \frac{x^2}{2} + \frac{u^2}{2}\right) = 0$$

using cauchy data

it is given that $u(x, x^{-1}) = 0$

--> $\text{ev}(\varphi, y = x^{-1}, u = 0) ;$
 $\text{ev}(\psi, y = x^{-1}, u = 0) ;$

$$x + \frac{1}{x} = c1$$

$$\frac{x^2}{2} + \frac{1}{2x^2} = c2$$

the relation is that $2c2+2=(c1)^2, (c1)^2-2c2=2$

--> $\text{ratsimp}((\text{lhs}(\varphi)^2) - 2 \cdot \text{lhs}(\psi) = 2) ;$

$$(2x + 2u)y + 2ux = 2$$

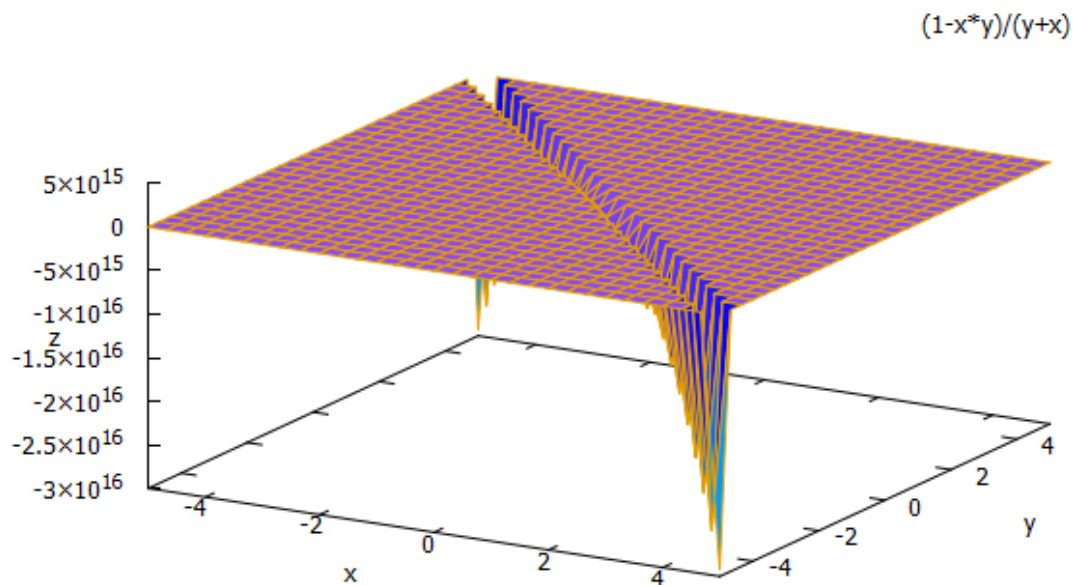
--> $\text{solve}((2 \cdot x + 2 \cdot u) \cdot y + 2 \cdot u \cdot x = 2, u) ;$

$$\left[u = -\frac{xy - 1}{y + x} \right]$$

--> $\text{plot3d}([(-(x \cdot y - 1) / (y + x)), [x, -5, 5], [y, -5, 5]]) ;$

[\"C : /Users/Student/AppData/Local/Temp/maxout4544.gnuplot\"]

Figure 3:



```
--> a ( x , y , u ) := 1 ;
      b ( x , y , u ) := - 1 ;
      c ( x , y , u ) := 1 ;
```

$$a(x, y, u) := 1$$

$$b(x, y, u) := -1$$

$$c(x, y, u) := 1$$

```
--> integrate ( 1 , x ) + integrate ( 1 , y ) ;
```

$$y + x$$

```
--> Φ : % = c1 ;
```

$$y + x = c1$$

```
--> integrate ( 1 , u ) + integrate ( 1 , y ) ;
```

$$y + u$$

```
--> Ψ : % = c2 ;
```

$$y + u = c2$$

```
--> f ( lhs ( Φ ) , lhs ( Ψ ) ) = 0 ;
```

$$f(y + x, y + u) = 0$$